

Abhilasha A. Kumar, Ph.D.

a.kumar@bowdoin.edu



[website](#)



[scholar](#)



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[github](#)

Academic Appointments

Assistant Professor , Psychology Bowdoin College, ME	2022 - present
Postdoctoral Research Associate Indiana University, Bloomington, IN	2021 - 2022

Education

Ph.D. in Psychological & Brain Sciences Washington University in St. Louis, MO	2021
M.A. in Psychological & Brain Sciences Washington University in St. Louis, MO	2018
M.A. in Liberal Arts (Computer Science and Psychology) Ashoka University, India	2016
Integrated M. Tech in Mathematics & Computing IIT Delhi, India	2014

External Funding

<i>“Collaborative Research: CompCog: Modeling Search within the Mental Lexicon”</i> National Science Foundation (BCS-PAC) Award No. 2235362, \$559,676 (Lead PI) Collaborative Grant w/ Michael Jones, Indiana University	2023-26
<i>“Integrating computational and neural methods to identify explore-exploit tradeoffs in memory search”</i> Maine IDeA Networks of Biomedical Research Excellence National Institutes of Health Recommended for funding [Pending]	2025-2026

Notable Awards and Honors

Fellow, Psychonomic Society	2025
Sabbatical Research Award Bowdoin College, \$10,000	2025
Faculty Research Award Bowdoin College, \$3,635	2023
Course Enrichment Award Bowdoin College, \$1600	2022, 2023

Manuscripts Under Review and in Preparation

 mentored postdoc  mentored student  conference paper  theoretical review

 Hambric, C. E.,  Sides, J., Holmes, K., **Kumar, A.A.** (under revision). Representing object state change during sentence comprehension: Boundary conditions and language-specific constraints.

 Hambric, C.E., Rammell, S.C., Kronenberger, W.G., Pisoni, D.B., Jones, M.N., **Kumar, A.A.** (under revision). Signatures of adaptive memory search: How early linguistic input shapes strategic use of lexical information.

 Hambric, C.E., Middleton, E., **Kumar, A.A.** (under review). Lexical Search and Retrieval in Aphasia: A Computational Analysis of Semantic Fluency.

Kumar, A.A., Steyvers, M. (in prep.). Pragmatic inference in helpful scenarios.

Peer-Reviewed Publications and Conference Proceedings

22.   Hambric, C.E., **Kumar, A.A.** (2025). Lexical search dynamics in taxonomic, thematic, and ad hoc categories. In *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (Vol. 45)*.
21.   Hambric, C.E., Holmes, K.J., **Kumar, A.A.** (2025). Are object state changes represented during language comprehension? A non-replication and extension. In *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (Vol. 45)*.
20. **Kumar, A.A.**, Hawkins, R.D. (2025). Lexical search and social reasoning jointly explain communication in associative reference games. *Journal of Experimental Psychology: General*. Advance online publication. <https://doi.org/10.1037/xge0001750>.
19. **Kumar, A. A.**, Lundin, N. B., & Jones, M. N. (2025). What's in *my* cluster? Evaluating automated clustering methods to understand idiosyncratic search behavior in verbal fluency. *Journal of Memory and Language*, 141, 104606. DOI: <https://doi.org/10.1016/j.jml.2024.104606>.
18.   **Kumar, A.A.**,  Kang, M., Kronenberger, W.G., Jones, M.N., Pisoni, D. (2024). Structure and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society (Vol. 46)*. DOI: <https://escholarship.org/uc/item/7vn9q9hh>.
17. **Kumar, A. A.**, Apsel, M., Zhang, L.,  Xing, N., & Jones, M. N. (2024). *forager*: A Python package and web interface for modeling mental search. *Behavior Research Methods*, 56(6), 6332-6348. DOI: <https://link.springer.com/article/10.3758/s13428-023-02296-x>.
16.   **Kumar, A.A.**, Steyvers, M. (2023). Help me help you: A computational model for goal inference and action planning. In *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (Vol. 45)*.

Science Society (Vol. 45). DOI: <https://escholarship.org/uc/item/2rk2b4sm>.

15. Coretta, S., Casillas, J. V., Roessig, S., Franke, M., Ahn, B., Al-Hoorie, A. H., ... **Kumar, A.A.**,... & Roettger, T.B. (2023). Multidimensional signals and analytic flexibility: Estimating degrees of freedom in human speech analyses. *Advances in Methods and Practices in Psychological Sciences*. 6(3), 25152459231162570. DOI: <https://doi.org/10.1177/25152459231162567>.
14.  **Kumar, A.A.**, Lundin, N. B., Jones, M.N. (2022). Mouse-mole-vole: The inconspicuous benefits of phonology during retrieval from semantic memory. In *Proceedings of the 44th Annual Meeting of the Cognitive Science Society* (Vol. 44). DOI: <https://escholarship.org/uc/item/4164178j>.
13.  Apsel, M., **Kumar, A.A.**, Jones, M.N. (2022). Finding the right words: A computational model of cued lexical retrieval. In *Proceedings of the 44th Annual Meeting of the Cognitive Science Society* (Vol. 44). DOI: <https://escholarship.org/uc/item/4kt1362j>.
12.  Kovacs, C.J.,  Wilson, J., **Kumar, A.A.** (2022). Fast and frugal memory search for communication. In *Proceedings of the 44th Annual Meeting of the Cognitive Science Society* (Vol. 44). DOI: <https://escholarship.org/uc/item/3301p4cj>.
11. Rosen, A. C., Lavacot, J. A., Porter, I. M., Chao, S. Z., Bikson, M., Kumar, A. A., & Cardenas, V. A. (2022). TDCS in a patient with dreadlocks: Improvements in COVID-19 related verbal fluency dysfunction. *Brain Stimulation: Basic, Translational, and Clinical Research in Neuromodulation*, 15(1), 254-256. DOI: [https://www.brainstimjrnl.com/article/S1935-861X\(22\)00005-5/](https://www.brainstimjrnl.com/article/S1935-861X(22)00005-5/).
10.  **Kumar, A. A.**, Steyvers, M., & Balota, D. A. (2022). A critical review of network-based and distributional approaches to semantic memory structure and processes. *Topics in Cognitive Science*, 14(1), 54-77. DOI: <https://doi.org/10.1111/tops.12548>.
9.  **Kumar, A.A.**, Garg. K., Hawkins, R.D. (2021). Contextual flexibility guides efficient communication in a cooperative word game. In *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society* (Vol. 43). DOI: <https://escholarship.org/uc/item/92m138t3>.
8. **Kumar, A.A.**, Steyvers, M., Balota, D.A. (2021). Semantic memory search and retrieval in a novel cooperative word game: A comparison of associative and distributional semantic models. *Cognitive Science*, 45(10), e13053. DOI: <https://doi.org/10.1111/cogs.13053>.
7.  **Kumar, A. A.** (2021). Semantic memory: A review of methods, models, and current challenges. *Psychonomic Bulletin & Review*, 28(1), 40-80. DOI: <https://doi.org/10.3758/s13423-020-01792-x>.
6. **Kumar, A.A.**, Balota, D.A., Steyvers, M. (2020). Distant connectivity and multiple-step priming in large-scale semantic networks. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 46 (12), pp. 2261-2276. DOI: <https://psycnet.apa.org/2019-72838-001>.
5. **Kumar, A.A.**, Balota, D.A. (2020). Attempted prime retrieval is a double-edged sword: facilitation and disruption in repeated lexical retrieval. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 46(8), 1505–1532. DOI: <https://psycnet.apa.org/2020-15198-001>.
4.  **Kumar, A.A.**, Balota, D.A., Steyvers, M. (2019). Distant concept connectivity in

- network-based and spatial word representations. In *Proceedings of the 41st Annual Meeting of the Cognitive Science Society*, 1348-1354. DOI: <https://escholarship.org/uc/item/4tn149vv>.
3. ***Kumar, A.A.**, *Balota, D.A., Habbert, J., Scaltritti, M., Maddox, G.B. (2019). Converging semantic and phonological information in lexical retrieval and selection in young and older adults. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 45(12), 2267–2289. DOI: <https://psycnet.apa.org/2019-14554-001>. *shared first author
 2. **Kumar, A.A.**, Hangal, S., Rosen, A.C. (2019). Autobiographical recall of personally familiar names and temporal information in e-mails: an automatic analytic approach using e-mail communications. *Behavior Research Methods*, 51(4), 1510-1530. DOI: <https://doi.org/10.3758/s13428-018-1182-9>.
 1. Maddox, G.B., Balota, D.A., **Kumar, A.A.**, Millar, P.,  Churchill, L. (2019). The immediate benefits and long-term costs of briefly presented primes on episodic recollection. *Journal of Memory and Language*. 106, 77-94. DOI: <https://doi.org/10.1016/j.jml.2019.02.003>.

Presentations

Invited research talks

- Kumar, A.A.** (September 2025). Exploring form-meaning interactions in novel word learning and memory search.
- *Basque Center on Cognition, Brain, and Language (BCBL), Spain.*
 - *Brave New Word Seminar, University of Milano-Bicocca, Milan, Italy.*
- Kumar, A.A.** (February 2025). Lexical search and social inference within the mental lexicon. *Department of Psychology, University of Wisconsin-Madison.*
- Kumar, A.A.** (February 2024, virtual). Search at multiple scales: Insights from individuals, dyads, and groups. *National University of Singapore, Singapore.*
- Kumar, A.A.** (October 2023, virtual). What is in a cluster? *Cognitive and Comparative Psychology Group, Brooklyn College & CUNY, NY.*
- Kumar, A.A.** (June 2023, virtual). Search and inference in cooperative contexts. *Computational Cognitive Science Lab, Princeton University, NJ.*
- Kumar, A.A.** (April 2023, virtual). Modeling search, communication, and cooperation. *Department of Psychology, Claremont McKenna College, CA.*
- Kumar, A.A.** (November 2022). Search and communication within the mental lexicon. *Colby College.*
- Kumar, A.A.** (June 2022, virtual). Careers in cognitive science. *Ashoka University, India.*
- Kumar, A.A.** (April 2021, virtual). Computational models of semantic memory representation, search, and retrieval.” *Indiana University, IN.*
- Kumar, A.A.** (November 2020, virtual). Semantic memory: Representation, search, and retrieval. *Bowdoin College, ME.*
- Kumar, A.A.** (September 2020, Virtual). Computational models of semantic memory. The Stanford Language and Cognition Lab, *Stanford University, CA.*

Kumar, A.A. (October 2020, virtual). Combining structure and process models of semantic memory. Computational Behavioral Science Lab. *University of Pennsylvania*, PA.

Conference talks (lead speaker)

Kumar, A.A. (July 2024). Memory search at multiple scales: Insights from individuals, dyads, and groups, *Dynamics of Memory Search Symposium, 46th Annual Meeting of the Cognitive Science Society*, Rotterdam, Netherlands.

Kumar, A.A., Steyvers, M. (July 2023, Virtual). Help me help you: A computational model for goal inference and action planning. *45th Annual Meeting of the Cognitive Science Society*, Sydney, Australia.

Kumar, A.A., Lundin, N.B., Jones, M.N. (July 2023, Virtual). Mouse-mole-vole: The inconspicuous benefits of phonology during retrieval from semantic memory, *44th Annual Meeting of the Cognitive Science Society*, Toronto, Canada.

Kumar, A.A. (July 2021). Language Games: A Window into Complex Cognition. *Games for Intelligence Workshop, 43rd Annual Meeting of the Cognitive Science Society (Virtual Conference)*.

Kumar, A.A. (July 2021). Fostering international student participation in cognitive science, *International Cognitive Science Affinity Group, 43rd Annual Meeting of the Cognitive Science Society (Virtual Conference)*.

Kumar, A.A., Balota, D.A. (October 2019). Priming for Distant Concepts: Insights from Semantic Networks, Behavior, Brain, and Cognition Brownbag, *Washington University in St. Louis*, St. Louis, MO.

Kumar, A.A. (June 2019). What's in an Abstract? Institute for Public Health Summer Research Program, *Washington University in St. Louis*, St. Louis, MO.

Kumar, A.A. (April 2019). Distant Connectivity in three 5000-word Semantic Networks in English". *Midwestern Psychological Association*, Chicago, IL.

Kumar, A.A., Balota, D.A. (December 2018). Lexical Retrieval Inhibition from Semantically Related Retrieval Primes, Aging and Development Brownbag, *Washington University in St. Louis*, St. Louis, MO.

Kumar, A.A. (October 2018). That Word at the Tip of Your Tongue. Public Health and Aging Session, *McDonnell Academy 7th International Symposium*, Beijing, China.

Kumar, A.A., Balota, D.A. (May 2018). Network Path Length as a Measure of Semantic Distance in a 5000-word English Lexicon, Data Blitz, *Show Me State Conference on Cognition*, St. Louis, MO.

Kumar, A.A., Rosen, A. C., Hangal, S. (March 2017). Remembering Names from Emails: Cognitive Experiments with Life-Logs (CELL): A Novel Approach to Study Name Retrieval, *Personal Digital Archiving Conference*, Stanford University, Palo Alto, CA

Kumar, A.A., Rosen, A. C., Hangal, S. (July 2016). Cognitive Experiments on Life-Logs (CELL): A New Approach to Study Recall of Personally Familiar Proper-Names using Emails, *International Conference on Memory*, Budapest, Hungary.

Posters

 **Kumar, A.A.**,  Hambric, C. (November 2024). Task and resource constraints modulate memory search: Evidence from gamified experiments with individuals, groups, and dyads. *Psychonomic Society 65th Annual Meeting, Psychonomic Society, New York, NY.*

 Lim, R.,  Cooney, A.,  Hambric, C., **Kumar, A.A.** (November 2024). Group differences in lexical source use and clustering during memory search. *Psychonomic Society 65th Annual Meeting, Psychonomic Society, New York, NY.*

 Lim, R., **Kumar, A.A.** (October 2024). CompCog: Modeling search within the mental lexicon. *President's Research Symposium, Bowdoin College, Maine.*

 Cooney, A., **Kumar, A.A.** (October 2024). Structure-level lexical interactions in memory search across multiple domains: A case study of normal hearing individuals and those with cochlear implants. *President's Research Symposium, Bowdoin College, Maine.*

Kumar, A.A.,  Kang, M., Kronenberger, W.G., Jones, M.N., Pisoni, D. (2024). Structure and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing. *46th Annual Meeting of the Cognitive Science Society, Rotterdam.*

 Sides, J., **Kumar, A.A.** (November 2023). Revisiting the Mental Representation of Object Properties and State Change in Situation Models. *Psychonomic Society 64th Annual Meeting, Psychonomic Society, San Francisco, CA.*

Apsel, M., **Kumar, A.A.**, Jones, M.N. (July 2022). Finding the right words: A computational model of cued lexical retrieval. *44th Annual Meeting of the Cognitive Science Society, July 2022, Toronto.*

Kumar, A.A. (November 2021). Probing Memory Representations through an Interactive Language Game, *Trends in Psychology Summit 2021, Virtual.*

Kumar, A.A., Balota, D.A., (November 2021). Modeling Distributional Structure and Spreading Activation in Retrieval Tasks, *Psychonomic Society 62nd Annual Meeting, Psychonomic Society, Virtual.* **Awarded the Psychonomic Society Graduate Travel Award*

Kumar, A.A., Balota, D.A., (November 2020). Searching Semantic Memory for Conceptual Associates, *Psychonomic Society 61st Annual Meeting, Psychonomic Society, Virtual.*

Kumar, A.A., Balota, D.A., Steyvers, M. (November 2019). Evaluating Association Networks and Distributional Semantic Models in a Cooperative Word Game, *Psychonomic Society 60th Annual Meeting, Psychonomic Society, Montreal, Canada.*

 Albertina, E., **Kumar, A.A.**, Balota, D.A. (April 2019). Comparing Semantic Models using a Cooperative Word Game. *Undergraduate Research Symposium, Washington University in St. Louis, MO.*

 Mutchler, Z., **Kumar, A.A.**, Balota, D.A. (April 2019). The Benefits and Costs of Repeated Retrieval from Semantic Memory. *Undergraduate Research Symposium, Washington*

University in St. Louis, MO.

Kumar, A.A., Balota, D.A., Steyvers, M. (November 2018). Distant Semantic Network Connectivity and Semantic Priming: Evidence from a 5000-word Semantic Network, *Psychonomic Society 59th Annual Meeting, Psychonomic Society*, New Orleans, LA.



Albertina, E., **Kumar, A.A.**, Balota, D.A. (October 2018). Investigating Network Connectivity for Distant Concepts using a 5000-word Semantic Network in English. *Undergraduate Research Symposium, Washington University in St. Louis*, MO.



Steiner, R., **Kumar, A.A.**, Balota, D.A. (May 2018). Effects of Priming on Immediate and Delayed Episodic Memory Retrieval. *Mind, Brain, and Behavior Symposium, Washington University in St. Louis*, MO.

Kumar, A.A., Balota, D.A. (May 2018). Prime Retrieval Failure Disrupts Lexical Retrieval to Definitions in Younger and Older Adults, *2018 Cognitive Aging Conference*, Atlanta, GA.

Kumar, A.A., Balota, D.A., Habbert, J., Scaltritti, M., Maddox, G.B. (November 2017). Priming Lexical Retrieval to Definitions in Younger and Older Adults, *Psychonomic Society 58th Annual Meeting, Psychonomic Society*, Vancouver, British Columbia, Canada.

Kumar, A.A., Hangal, S., Piratla, V., Rosen, A.C. (July 2017). Whom Did I Send This Message To: An Automatic Analytic Approach to 59000 Email Communications, *2017 Leading Edge Workshop: Beyond the Lab: Using Big Data to Discover Principles of Cognition, Psychonomic Society*, Madison, WI.

Teaching Experience

Instructor, Bowdoin College

Data Analysis (PSYC 2520)

Spring 2024

Cognition: Methods and Models (PSYC 2040)

Spring 2023, Spring 2024

Laboratory in Cognitive Science (PSYC 2740)

Fall 2022, Fall 2023, Fall 2024

Intelligent Minds and Machine (PSYC 3043)

Fall 2022, Fall 2023, Fall 2024

Research mentoring and supervision, Bowdoin College

Summer 2025

Ethan Lam – '27 Psychology and Mathematics major

Kyra Brown – '28 Psychology and Mathematics major

Anushka Elavia – '27 Biochemistry major and Sociology minor

Spring 2025

Ethan Lam – '27 Psychology and Mathematics major (Independent Study)

Summer 2024

Amanda Cooney – '26 Psychology major and Computer Science minor (NSF Research Fellow)

Richard Lim – '26 Prospective Computer Science and Mathematics major (Gibbons Summer Research Fellow)

Spring 2024

William Warlick – '24 Computer Science and Philosophy major (NSF Research Fellow)
 Mingi Kang – '26 Computer Science and Mathematics major (NSF Research Fellow)

Fall 2023

Stephanie Chun – '24 Neuroscience major, Asian Studies minor (Independent Study)
 Ulemj Munkhtur – '26 Computer Science and Economics major (Independent Study)
 Carrie Zou – '24 Psychology and Economics major (Independent Study)
 Simon Thomas – '25 Physics and Hispanic Studies major (Research Assistant)
 Mingi Kang – '26 Computer Science and Mathematics major (Research Assistant)

Spring 2023

Thomas Mazzuchi – '23 Psychology and Mathematics major (Honors)
 Reed Warburton – '23 Psychology and Biology major (Independent Study)
 Kavya Doraswamy – '24 Psychology and Computer Science major (Independent Study)
 Jon Sides – '24 Psychology and Mathematics major (Independent Study)
 Mingi Kang – '26 Computer Science and Mathematics major (Independent Study)
 Nancy Xing – '24 Computer Science major, Psychology minor (Research Assistant)

Fall 2022

Thomas Mazzuchi '23 Psychology and Mathematics major (Honors)
 Reed Warburton '23 Psychology and Biology major (Independent Study)

Pedagogical Development Workshops

Faculty Fellows Program, Bowdoin College	2024-2025
Course Redesign Institute, Bowdoin College	August 2024
Colby, Bates, and Bowdoin Pedagogy Matters Conference: The Cognitive Challenges of Effective Teaching	August 2024
Inclusivity and Social Annotation: Fostering Diverse Learning Environments, Hypothes.is	April 2023
Colby, Bates, and Bowdoin Pedagogy Matters Conference: The Call Is Coming from Inside the House: How Grades Undermine Learning and Jeopardize Student Wellbeing	August 2023
Ideas for Effective Groupwork with Whitney Hogan and Katie Byrnes	December 2022
Colby, Bates, and Bowdoin Pedagogy Matters Conference: Each Day is a New Labor: What Can Black Campus Life Teach Us About Pedagogy?	August 2022
Mentoring Undergraduate Students, National Research Mentoring Network	July 2022

Guest Lecturer, Washington University in St. Louis

Hierarchical Linear Models [graduate class]	Spring 2019
Methods in Linguistics Research	Fall 2018
Introduction to Computational Linguistics	Spring 2018
Introduction to Psychological Statistics	Spring 2018
Introduction to Psychology	Fall 2017

Assistant to the Instructor, Washington University in St. Louis

Psychology of Language	Spring 2020
Hierarchical Linear Models [graduate class]	Spring 2019
<i>*Awarded the Outstanding Assistant to Instructor Award</i>	
Methods in Linguistics Research	Fall 2018
Introduction to Psychological Statistics	Spring 2018
Introduction to Psychology	Fall 2017

Research Mentoring, Washington University in St. Louis

Students and Teachers as Research Scientists [STARS program, high-school students]	Summer 2017
Cognitive Psychology Lab	2016 – 2021

Teaching Assistant, Ashoka University, India

Probability, Statistics & Data Science	Spring 2016
Human-Computer Interaction	Fall 2015
Mathematical Thinking	Spring 2014

Professional Development in Teaching Program, Washington University in St. Louis

Completed a formalized training program in pedagogy involving 3 mentored experiences
 Attended advanced workshops on effective evidence-based teaching practices:

Philosophy and Implementation of Collaborative Learning	January 2020
Applying Cognitive Science to Improve Teaching	April 2019
Providing Verbal Feedback to Students	February 2019
Who's in Charge Here? Strategies for Managing a Classroom	August 2017

Professional Service

Committee on Teaching and Classroom Practice (CoTCP) <i>Bowdoin College</i>	2024 – 2025
Mentoring Committee <i>Cognitive Science Society</i>	2024 –
Organizing team <i>Advancing Cognitive Research with Undergraduates (ACRU)</i>	2024 –

International Committee
Cognitive Science Society

2022 – 2024

Editorial Board Service

Editorial Fellow
Journal of Experimental Psychology:
Learning, Memory, and Cognition

2023 – 2024

Ad Hoc Reviewer

Acta Psychologica
 Behavior Research Methods
 Canadian Journal of Experimental Psychology
 Cognition
 Cognitive Processing
 Cognitive Science
 Cognitive Science Society Annual Conference Proceedings
 Consciousness
 Cortex
 Communications Psychology
 Frontiers in Artificial Intelligence
 Journal of Cognitive Psychology
 Journal of Experimental Psychology: General
 Journal of Experimental Psychology: Human Perception and Performance
 Journal of Experimental Psychology: Learning, Memory, and Cognition
 Journal of Memory and Language
 Memory and Cognition
 Nature Reviews Psychology
 National Science Foundation
 Neuropsychologia
 Psychological Research
 Psychological Science
 Psychological Review
 Psychonomic Bulletin and Review
 Topics in Cognitive Science

Memberships in Professional Organizations

Cognitive Science Society (CSS)
 Psychonomic Society
 SPARK Society
 Women in Cognitive Science (WiCS)
 Women in Network Science (WiNS)
 Graduate Women in Science